数理分析方法

连续函数

Lecture 107

原课件逐页保留 · 零基础中文讲解 · 证明训练

材料与编写说明

课程原页：张顺明教授课件（用户提供）。参考：Gérard Debreu, Theory of Value (1959)，第一章 Mathematics。原页与本讲义的补充、纠错、例题分开呈现。本文为辅助自学材料，不代表教师官方讲义。

怎样阅读

先看本页的中文说明，再对照原页符号；遇到证明，先抄出假设和目标，再逐行说明“这一步用的是什么”。遮住证明重写一遍，最后用反例检查条件是否必要。重复或动画展开页仍逐页保留；说明会指出新增步骤或对应的完整论证位置。

“补充”用于补基础或展开原课件略去的论证；“纠错”指出原页需要修正的地方。课程概览及人物介绍页按其原有用途解释，不虚构数学定理。

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零基础读证明：随手查阅

对象的层次。 $x \in A$ 表示一个元素属于集合，$A \subseteq B$ 表示 A 的每个元素都在 B 中。$\{x\}$ 是只含 x 的集合，不是 x 本身。$\varnothing$ 是空集，$\{\varnothing\}$ 则含有一个元素。$\mathbb{R}^N$ 中的点是有 N 个坐标的有序列表。

量词。 $\forall$ 是“对所有”，$\exists$ 是“至少存在一个”。例如“每个学生有一本书”不要求是同一本；“有一本书供每个学生使用”则要求同一本。证明极限时 $\forall \varepsilon > 0\ \exists n_0\ \forall n \geq n_0$ 的顺序不可换：先指定精度，再找统一起点，最后控制这个起点之后的全部项。

怎样否定。 否定“所有 x 满足 P”得到“存在 x 不满足 P”；否定“存在”得到“所有都不”。否定 $P \Rightarrow Q$ 是 P 真且 Q 假。因此要推翻一个定理，必须满足其全部假设又违反结论；拿不满足假设的例子不能当反例。

四种基本证明。 (1) 包含 $A \subseteq B$：任取 $x \in A$，用定义得到 $x \in B$。(2) 相等 $A = B$：两个方向分别证明。(3) 等价 $P \iff Q$：证明 $P \Rightarrow Q$ 及反向。(4) 反证：把结论否定加入已知，推出矛盾。逆否命题 $\neg Q \Rightarrow \neg P$ 与 $P \Rightarrow Q$ 等价，“反过来” $Q \Rightarrow P$ 则未必成立。

数列、子列和收敛。 数列 x_n 是第 n 个对象；子列 x_{n_k} 的指标必须严格递增。$x_n \to x$ 指任意小的误差最终都能保持。前面有限多项不影响极限。抽子列通常是为了用有界性和有限维 Bolzano–Weierstrass 定理获得极限；抽取之后须把原来所有相关量一起限制到同一子列。

距离的关键估计。 $\|x\| = (\sum_i x_i^2)^{1/2}$，$\|x - y\|$ 是距离。三角不等式 $\|x - z\| \leq \|x - y\| + \|y - z\|$ 允许经中间点拆误差。两段误差各小于 $\varepsilon/2$，总误差便小于 ε；分成 m 段时可各分配 ε/m。这里的 ε 不是固定的小数，而是任意正精度。

做证明题的五行草稿。 先写完整假设；写准确目标；列出可用定义与定理；逐步注明所用条件；最后检查任意性、存在性和量词顺序。出现“显然”时问：若阅卷者要求展开，我能否补出一个集合等价链、一个不等式或一个明确构造？不知道证明的命题要标为引用，不能把待证结论循环当作理由。

1 原课件第 1 页：连续性在本课程中的作用

Theory of Value

An Axiomatic Analysis of Economic Equilibrium

Gerard Debreu (1959)

Professor Shunming Zhang

School of Finance
Renmin University of China

September 26, 2022

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

函数连续描述输入小变动不会造成输出的大跳跃。本讲从数列判据开始，逐步得到紧集上的极值与连通集上的介值，再研究只有一个方向受控制的半连续性。

2 原课件第 2 页：连续性在本课程中的作用

Chapter 1: Mathematics

01. Introduction
02. Sets
03. Functions and Correspondences
04. Preorderings
05. Real Numbers
06. Limits in $\mathbb{R}^N$
07. Continuous Functions
08. Continuous Correspondences
09. Vectors in $\mathbb{R}^N$
10. Fixed Points

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“连续性在本课程中的作用”。完整定义、条件和论证见本讲原页第 1 页下方。

3 原课件第 3 页：序列连续与复合、坐标

Continuous Functions

In this section, S denotes a subset of $\mathbb{R}^K$, T a subset of $\mathbb{R}^L$, (x^q) a sequence of points of S, and (y^q) a sequence of points of T.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

设 $f : S \to T$，在 x 连续指每个 $x_n \in S, x_n \to x$ 都满足 $f(x_n) \to f(x)$。复合连续：先由 f 连续得 $f(x_n) \to f(x)$，再由 g 连续取像。有限维中 $\|z_n - z\| \to 0$ 等价于每个坐标收敛：一边用 $|z_{ni} - z_i| \le \|z_n - z\|$，另一边用有限和 $\|z_n - z\|^2 = \sum_i |z_{ni} - z_i|^2$。所以向量函数连续当且仅各坐标函数连续。

4 原课件第 4 页：序列连续与复合、坐标

Continuous Functions

Let f be a function from S to T, and consider a point $x^0 \in S$. The function f is continuous at the point x^0 if:

$$\text{“}x^q \to x^0, y^0 = f\left(x^0\right), y^q = f\left(x^q\right)\text{”} \text{ implies } \text{“}y^q \to y^0\text{”}$$

In other words. f is continuous at x^0 if the image of any sequence tending to x^0 is a sequence tending to the image of x^0.

Example: Let f be the real-valued function of a real variable (i.e., a function from $\mathbb{R}$ to $\mathbb{R}$) defined by $y = \dfrac{1}{x}$ if $x \neq 0$, and $y = 0$ if $x = 0$. The function f is continuous at every point of $\mathbb{R}$, with the exception of 0, as its graph readily suggests.

The function f is continuous on S if it is continuous at every point of S.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“序列连续与复合、坐标”。完整定义、条件和论证见本讲原页第 3 页下方。

5 原课件第 5 页：序列连续与复合、坐标

Continuous Functions

Let S_1, S_2, T be subsets of $\mathbb{R}^{K_1}$, $\mathbb{R}^{K_2}$, $\mathbb{R}^{L}$ respectively, f be a function from S_1 to S_2 and g be a function from S_2 to T. Define a function h from S_1 to T by $h(x) = g(f(x))$ for every x in S_1. It is immediate that:

(1) If f is continuous at the point x of S_1, and if g is continuous at the point $f(x)$ of S_2, then h is continuous at x.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“序列连续与复合、坐标”。完整定义、条件和论证见本讲原页第 3 页下方。

6 原课件第 6 页：序列连续与复合、坐标

Continuous Functions

For every $n = 1, 2, \cdots, N$, let T_n be a subset of $\mathbb{R}^{L_n}$ and consider the product $T = \prod_{n=1}^{N} T_n$. It is immediate that the projection on T_n is continuous on T.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“序列连续与复合、坐标”。完整定义、条件和论证见本讲原页第 3 页下方。

7 原课件第 7 页：序列连续与复合、坐标

Continuous Functions

With the notations of the last paragraph, let f_n be a function from S to T_n, and define a function f from S to T by $f(x) = (f_n(x))$ for every x in S, where $(f_n(x))$ denotes the N-tuple $(f_1(x), f_2(x), \cdots, f_N(x))$. It is immediate that:

(2) If every f_n is continuous at the point x of S, then f is continuous at x.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“序列连续与复合、坐标”。完整定义、条件和论证见本讲原页第 3 页下方。

8 原课件第 8 页：连续性的五个等价条件

Continuous Functions

One can prove:

(3) A function f from S to T is continuous on S if and only if the inverse image of every set closed in T is closed in S.

Proposition: A mapping f from a metric space X to a metric space Y. The following statements are equivalent:

(1) f is continuous.

(2) For each open set G in Y, the set $f^{-1}(G)$ is an open set in X.

(3) For each closed set F in Y, the set $f^{-1}(F)$ is an closed set in X.

(4) For every set A in X, $f(\overline{A}) \subseteq \overline{f(A)}$.

(5) For every set B in Y, $\overline{f^{-1}(B)} \subseteq f^{-1}(\overline{B})$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

连续的 ε–δ 形式为：对每个 $\varepsilon > 0$ 存在 $\delta > 0$，所有定义域内距 x 小于 δ 的点，其像距 $f(x)$ 小于 ε。否定它可逐个取 $\delta = 1/n$ 构造 $x_n \to x$ 而像距始终至少 ε_0，所以与数列定义等价。连续推出开集逆像开：$f(x) \in G$ 先取像球包含于 G，再用连续性拉回一个输入球。开逆像与闭逆像用补集对偶。闭逆像推出 $f(\overline{A}) \subseteq \overline{f(A)}$：后者的逆像是含 A 的闭集，故含闭包。反向若有闭包包含性质，取 $A = f^{-1}(F)$ 且 F 闭，则 $f(\overline{A}) \subseteq F$，故 $\overline{A} \subseteq A$，逆像闭。不要把像闭包上的横线漏掉。

9 原课件第 9 页：连续性的五个等价条件

Continuous Functions

Proof: The function f is continuous at the point x_0 if for any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\|x - x_0\| < \delta \qquad \text{implies} \qquad \|f(x) - f(x_0)\| < \varepsilon,$$

or, the function f is continuous at the point x_0 if for any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x \in N_\delta(x_0) \qquad \text{implies} \qquad f(x) \in N_\varepsilon(f(x_0)),$$

that is, $f(N_\delta(x_0)) \subseteq N_\varepsilon(f(x_0))$, equivalently, the function f is continuous at the point x_0 if for any neighborhood V of $f(x_0)$, there exists neighborhood U of x_0 such that $f(U) \subseteq V$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“连续性的五个等价条件”。完整定义、条件和论证见本讲原页第 8 页下方。

10 原课件第 10 页：连续性的五个等价条件

Continuous Functions

Proof of (1) $\Longrightarrow$ (2): For each open set G in Y, we want to check that set $f^{-1}(G)$ is an open set in X.
For any $x_0 \in f^{-1}(G)$, $f(x_0) \in G$. For any neighborhood V of $f(x_0)$ with $f(x_0) \in V \subseteq G$, there exists neighborhood U of x_0 such that $f(U) \subseteq V$, then $x_0 \in U \subseteq f^{-1}(V) \subseteq f^{-1}(G)$.
Proof of (2) $\Longrightarrow$ (1): For any neighborhood V of $f(x_0)$, there exists neighborhood V' of $f(x_0)$ such that $f(x_0) \in V' \subseteq V$, then $x_0 \in f^{-1}(V') \subseteq f^{-1}(V)$ and $f^{-1}(V')$ is an open set.
Proof of (2) $\Longrightarrow$ (3): For each closed set F in Y, $G \equiv Y \setminus F$ is open, then
$f^{-1}(G) = f^{-1}(Y \setminus F) = f^{-1}(Y) \setminus f^{-1}(F) = X \setminus f^{-1}(F)$ is open, then the set $f^{-1}(F)$ is an closed set in X.
Proof of (3) $\Longrightarrow$ (2): For each open set G in Y, $F \equiv Y \setminus G$ is closed, then
$f^{-1}(F) = f^{-1}(Y \setminus G) = f^{-1}(Y) \setminus f^{-1}(G) = X \setminus f^{-1}(G)$ is closed, then the set $f^{-1}(G)$ is an open set in X.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“连续性的五个等价条件”。完整定义、条件和论证见本讲原页第 8 页下方。

11 原课件第 11 页：连续性的五个等价条件

Continuous Functions

Proof of (3) $\Longrightarrow$ (4): For every set A in X, $f(A) \subseteq \overline{f(A)}$, then $A \subseteq f^{-1}\left(\overline{f(A)}\right)$ and $f^{-1}\left(\overline{f(A)}\right)$ is a closed set in X, hence $\overline{A} \subseteq f^{-1}\left(\overline{f(A)}\right)$. Thus $f\left(\overline{A}\right) \subseteq \overline{f(A)}$.

Proof of (4) $\Longrightarrow$ (3): For each closed set F in Y, define $A = f^{-1}(F)$, then $f(A) = f\left(f^{-1}(F)\right) \subseteq F$, hence $f\left(\overline{A}\right) \subseteq \overline{f(A)} \subseteq \overline{F} = F$. Thus $\overline{A} \subseteq f^{-1}(F) = A$, $A = \overline{A}$ is closed, that is, the set $f^{-1}(F)$ is an closed set in X.

Proof of (3) $\Longrightarrow$ (5): For every set B in Y, $f^{-1}(B) \subseteq f^{-1}\left(\overline{B}\right)$ and $f^{-1}\left(\overline{B}\right)$ is a closed set in X, hence $\overline{f^{-1}(B)} \subseteq f^{-1}\left(\overline{B}\right)$.

Proof of (5) $\Longrightarrow$ (3): For each closed set F in Y, $\overline{f^{-1}(F)} \subseteq f^{-1}\left(\overline{F}\right) = f^{-1}(F)$, then the set $f^{-1}(F)$ is an closed set in X.

Proof of (4) $\Longrightarrow$ (5): For every set B in Y, $f^{-1}(B) \subseteq X$, then $f\left(\overline{f^{-1}(B)}\right) \subseteq \overline{f\left(f^{-1}(B)\right)} \subseteq \overline{B}$, hence $\overline{f^{-1}(B)} \subseteq f^{-1}\left(\overline{B}\right)$.

Proof of (5) $\Longrightarrow$ (4): For every set A in X, $f(A) \subseteq Y$, then $\overline{A} \subseteq \overline{f^{-1}(f(A))} \subseteq f^{-1}\left(\overline{f(A)}\right)$, hence $f\left(\overline{A}\right) \subseteq \overline{f(A)}$. $\square$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“连续性的五个等价条件”。完整定义、条件和论证见本讲原页第 8 页下方。

12 原课件第 12 页：实值连续函数的水平集

Continuous Functions

In the important particular case where f is a real-valued function on S. i.e., where f is a function from S to $\mathbb{R}$, one can prove:

(3′) A function from S to $\mathbb{R}$ is continuous on S if and only if the inverse image of every interval of $\mathbb{R}$ of the form $(-\infty, y]$ or $[y, \infty)$ is closed in S.

Example: (1) Let $\mathbb{R}_+$ be the set of positive reals, and consider the function f from $S = \mathbb{R} \times \mathbb{R}_+$ to $\mathbb{R}$ defined by $y = f(u, v) = \frac{u}{v}$. "The function f is continuous on S" is equivalent to "the inverse image of $(-\infty, y]$, i.e, the set $\{(u, v) \in S \mid u - yv \leqq 0\}$, is closed in S, and similarly for $[y, \infty)$; the second assertion is readily suggested by a drawing in $\mathbb{R}^2$, and easily proved. (2) The example of 1.7.b does not satisfy the continuity criterion of (3′), as the graph shows.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

实值函数连续当且仅所有 $\{f \leq a\}$ 与 $\{f \geq a\}$ 闭。必要性是闭集逆像；充分性先取补得 $\{f > a\}, \{f < b\}$ 开，再相交得 $f^{-1}((a,b))$ 开，任意实开集为开区间并，所以逆像开。实开集的极大区间分解见106第53页：不交区间各含不同有理数，因此可数。

13 原课件第 13 页：实值连续函数的水平集

Continuous Functions

Proof: If A function from S to $\mathbb{R}$ is continuous on S, as every interval of $\mathbb{R}$ of the form $(-\infty, y]$ or $[y, \infty)$ is closed in $\mathbb{R}$, then the inverse image is also closed in S.

Conversely, if the inverse image of every interval of $\mathbb{R}$ of the form $(-\infty, y]$ or $[y, \infty)$ is closed in S. Assume the function is $f : S \to \mathbb{R}$. So for any open interval of $\mathbb{R}$, the form (a, b), (a, ∞), $(-\infty, a)$, we have

$$f^{-1}(-\infty, a) = f^{-1}(\mathbb{R}) \setminus f^{-1}[a, \infty)$$
$$f^{-1}(a, \infty) = f^{-1}(\mathbb{R}) \setminus f^{-1}(-\infty, a]$$
$$f^{-1}(a, b) = f^{-1}(-\infty, b) \setminus f^{-1}(-\infty, a]$$

are all open in S.

We also have following lemma: Every open set in $\mathbb{R}$ is the countable union of disjoint open intervals. So for any open set $O \subseteq \mathbb{R}$, there exists countable open interval $\{I_n; n \in \mathbb{N}\}$ such that $O = \bigcup_{n=1}^{\infty} I_n$. So

$$f^{-1}(O) = \bigcup_{n=1}^{\infty} f^{-1}(I_n)$$

are also open in S. Hence f is continuous. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“实值连续函数的水平集”。完整定义、条件和论证见本讲原页第 12 页下方。

14 原课件第 14 页：实值连续函数的水平集

Continuous Functions

Lemma: Every open set in $\mathbb{R}$ is the countable union of disjoint open intervals.

Proof: Let G be a non-empty open set in $\mathbb{R}$. Write $a \sim b$ if the closed interval $[a, b]$ if $a < b$ or $[b, a]$ if $b < a$, lies in G. This is an equivalence relation, in particular $a \sim b$ since $\{a\}$ is itself a closed interval. G is therefore the union of disjoint equivalence classes. Let $C(a)$ be the equivalence class containing a. Let $G = \bigcup_{a \in I} C(a)$ where $C(a) \neq C(a')$ if $a \neq a'$. $C(a)$ is clearly an interval. Also $C(a)$ is open, for if $k \in C(a)$, then $(k - \varepsilon, k + \varepsilon) \subseteq G$ for sufficiently small ε. But then $(k - \varepsilon, k + \varepsilon) \subseteq C(a)$; so G is the union of disjoint intervals. For each $C(a)$ choose a rational q_a, this map $I \to \mathbb{Q} : G(a) \longmapsto q_a$ is injective, so I is countable. $\square$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“实值连续函数的水平集”。完整定义、条件和论证见本讲原页第 12 页下方。

15 原课件第 15 页：紧集的连续像与极值

Continuous Functions

One can prove:

(4) Let f be a function from S to T. If f is continuous on S, and if S is compact, then $f(S)$ is compact.

Proof: Let $\bigcup_{\gamma\in\Gamma} U_\gamma$ be an open covering of $f(S)$ on T. Then the sets $f^{-1}(U_\gamma)$ are open and $\left\{f^{-1}(U_\gamma);\gamma\in\Gamma\right\}$ form an open covering of S since we have show

$$f^{-1}\left(\bigcup_{\gamma\in\Gamma} U_\gamma\right)=\bigcup_{\gamma\in\Gamma} f^{-1}(U_\gamma)$$

Since S is compact, there exists a finite subset $\Gamma'\subseteq\Gamma$ such that $\bigcup_{\gamma\in\Gamma'} f^{-1}(U_\gamma)$ still forms a covering of S, then

$$f(S)=f\left(\bigcup_{\gamma\in\Gamma'} f^{-1}(U_\gamma)\right)=f\left(f^{-1}\left(\bigcup_{\gamma\in\Gamma'} U_\gamma\right)\right)\subseteq\bigcup_{\gamma\in\Gamma'} U_\gamma$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

设 K 非空紧，$\{U_i\}$ 覆盖 $f(K)$，则 $\{f^{-1}(U_i)\}$ 为 K 的开覆盖。取有限子覆盖再取像得 $f(K)$ 的有限子覆盖，故像紧。实值时像非空闭有界，令 $M=\sup f(K)$，选像中数 $y_n>M-1/n$，闭性给 $M\in f(K)$，故有 $x_M\in K$ 达到最大。最小同理。M 是值，x_M 是达到该值的点；最优点可以不唯一。

16 原课件第 16 页：紧集的连续像与极值

Continuous Functions

Applying this result to the particular case where f is real-valued, one obtains immediately:

(4′) (Weierstrass' Theorem) Let f be a function from S to $\mathbb{R}$. If f is continuous on S, and if S is non-empty compact, then $f(S)$ has a maximum and a minimum.

Proof: $f(S)$ is a compact subset of $\mathbb{R}$, hence closed and bounded. Now, any bounded set $A \subseteq \mathbb{R}$ has a least upper bound $\sup A$ and a greatest lower bound $\inf A$, and these two points belong to the closure A. But applying this to $A = f(S)$, which is closed, we conclude that $\sup f(S)$ and $\inf f(S)$ belong to $f(S)$ itself, which is exactly what is being claimed. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“紧集的连续像与极值”。完整定义、条件和论证见本讲原页第 15 页下方。

17 原课件第 17 页：紧集的连续像与极值

Continuous Functions

"$f(S)$ has a maximum" also means (1.5.g) "there is an x^M in S such that for all $x \in S$ one has $f(x) \leqq f(x^M)$." Such an element is called a maximizer of f. "$f(S)$ has a minimum" also means (1.5.g) "there is an x^m in S such that for all $x \in S$ one has $f(x^m) \leqq f(x)$." Such an element is called a minimizer of f. The maximum (resp. minimum) of the set $f(S)$ is also called the maximum (resp. minimum) of the function f on S.

Example: Let f be the function defined in the example of 1.7.b, and I an interval to which the variable x in restricted. One sees on the graph of f how, when I is not bounded, or not closed, or owns 0 and a different point, f has no maximizer or no minimizer.

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本页继续“紧集的连续像与极值”。完整定义、条件和论证见本讲原页第 15 页下方。

18 原课件第 18 页：连通像与介值

Continuous Functions

It is easy to prove (using (3)),

(5) Let f be a function from S to T. If f is continuous on S, and if S is connected, then $f(S)$ is connected.

Proof: If $f(S)$ is not connected in Y, then we can find two non-empty and disjoint open subsets V_1 and V_2 of $f(X)$ such that $V_1 \cup V_2 = f(S)$. But then, $f^{-1}(V_1)$ and $f^{-1}(V_2)$ are nonempty open subsets of S. Moreover, these sets are disjoint, and we have $f^{-1}(V_1) \cup f^{-1}(V_2) = f^{-1}(V_1 \cup V_2) = S$. Thus S is not connected. Contradiction! □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若 $f(K) = A \cup B$ 是两个非空不交相对开集的分离，则逆像为 K 的非空不交相对开分离，矛盾。实数连通集是区间，所以连续函数的两个函数值之间的每个数也是某个函数值。在 $[a,b]$ 上，若 $f(a)f(b) < 0$ 就存在零点。仅能得存在，不保证唯一；唯一通常还需严格单调等条件。

19 原课件第 19 页：连通像与介值

Continuous Functions

Applying this result to the particular case where f is real-valued, one obtains immediately (see (8) of 1.6.u):

(5′) (Bolzano's Theorem) Let f be a function from S to $\mathbb{R}$. If f is continuous on S, and if S is connected, then $f(S)$ is an interval.

Thus, if x^1, x^2 are two points of S, and y is a real number such that $f\left(x^1\right) \leqq y \leqq f\left(x^2\right)$, then there is an $x \in S$ such that $f(x) = y$. In other words, f takes on all values between $f\left(x^1\right)$ and $f\left(x^2\right)$. This result is often applied to the case where S is an interval of $\mathbb{R}$.

The Intermediate Value Theorem. Let X be a connected metric space and $f : X \to \mathbb{R}$ a continuous function. If $f(x_1) \leqslant \alpha \leqslant f(x_2)$ for some $x_1 \in X$ and $x_2 \in X$, then there exists a $x \in X$ such that $f(x) = \alpha$.

Proof. $f(X)$ is connected in $\mathbb{R}$. Hence $f(X)$ must be an interval. □

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本页继续“连通像与介值”。完整定义、条件和论证见本讲原页第 18 页下方。

20 原课件第 20 页：邻域语言与量词

Notions of Continuity for Functions

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$ be a function. When it is clear to which space a pair of points belongs, I may drop the subscripts on d_X and d_Y and simply write d for the metric. I will also use the notion $B_\varepsilon(x)$ to denote the open ball of radius ε centered at x when the metric is obvious.

Recall that the Euclidean metric on $\mathbb{R}^N$ (or l^2) is given by

$$d_2\left(x^1, x^2\right) = \sqrt{\sum_{n=1}^{N} \left(x_n^1 - x_n^2\right)^2}$$

Other useful metrics on $\mathbb{R}^N$ include the max (or l^∞) metric

$$d_\infty\left(x^1, x^2\right) = \max\left\{\left|x_n^1 - x_n^2\right| : 1 \leqslant n \leqslant N\right\}$$

and the taxicab (or l^1) metric

$$d_1\left(x^1, x^2\right) = \sum_{n=1}^{N} \left|x_n^1 - x_n^2\right|$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

邻域包含一个以该点为中心的开球。连续要求每个输出邻域的逆像都是输入点的邻域，不要求逆像本身是球。量词顺序为 $\forall \varepsilon\ \exists \delta\ \forall z$，允许 δ 依赖 x 与 ε，不能在看到 z 后再选它。例 $f(t) = t^2$ 在 x 附近取 $\delta = \min(1, \varepsilon/(2|x|+1))$，因 $|z^2 - x^2| \le |z-x|(2|x|+1)$。

21 原课件第 21 页：邻域语言与量词

Notions of Continuity for Functions

Definition 1 (Continuity using neighborhoods). A function $f : X \to Y$ is continuous at x if for every neighborhood V of $f(x)$, its inverse image $f^{-1}(V)$ is a neighborhood of x. That is, for every open set V with $f(x) \in V$, there exists an open set U with $x \in U$ such that

$$z \in U \Longrightarrow f(z) \in V,$$

i.e., $f(U) \subseteq V$.

Note that this definition is topological, that is, it uses only the notion of neighborhoods and does not mention the metrics. The next definition makes explicit use of the metrics.

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本页继续“邻域语言与量词”。完整定义、条件和论证见本讲原页第 20 页下方。

22 原课件第 22 页：邻域语言与量词

Notions of Continuity for Functions

Definition 2 (Continuity using metrics). A function $f : X \to Y$ is continuous at x if for every $\varepsilon > 0$ there exists some $\delta > 0$ such that

$$d_X(x, z) < \delta \Longrightarrow d_Y(f(x), f(z)) < \varepsilon.$$

Definition 3 (Continuity using sequences). A function $f : X \to Y$ is continuous at x if whenever $x_n \to x$, then $f(x_n) \to f(x)$.

Proposition 1. A function f is continuous at x in the topological sense of Definition 1 if and only if it is continuous at x in the metric sense of Definition 2 if and only if it is continuous at x in the sequential sense of Definition 3.

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本页继续“邻域语言与量词”。完整定义、条件和论证见本讲原页第 20 页下方。

23 原课件第 23 页：实值半连续的定义与等价判据

Numerical Functions Defined on a Topological Space

The ideas of continuity introduced above apply in particular to numerical functions and we say that a numerical function f, defined on a topological space X, is continuous at a point x_0 if it is a mapping of X into $\mathbb{R}$ and, as such, is continuous at x_0; in other words if, to each $\varepsilon > 0$, there corresponds a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad |f(x) - f(x_0)| < \varepsilon.$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

下半连续指 $f(x) \le \liminf f(x_n)$，允许向下的低点；上半连续指 $f(x) \ge \limsup f(x_n)$，允许向上的高点。下半连续等价于 $\{f > a\}$ 开，也等价于 $\{f \le a\}$ 闭：若 $a < f(x)$，定义给邻域中 $f > a$；反向对任意 $a < f(x)$ 用该开集取邻域。上半连续用 $-f$ 转化，得到 $\{f < a\}$ 开及 $\{f \ge a\}$ 闭。两者同时成立即连续。例 $f(0) = 0, f(x) = 1$ 对 $x \ne 0$ 下半连续但不连续，其负函数上半连续。

24 原课件第 24 页：实值半连续的定义与等价判据

Numerical Functions Defined on a Topological Space

In certain situations this idea turns out to be too restrictive and we are forced to consider the concepts of lower and upper semi-continuity, as given in the following definitions.

A numerical function f defined on X is said to be lower semi-continuous (l.s.c.) at a point x_0 if, for each $\varepsilon > 0$, there corresponds a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad f(x) > f(x_0) - \varepsilon.$$

A numerical function f defined on X is said to be upper semi-continuous (u.s.c.) at a point x_0 if, for each $\varepsilon > 0$, there corresponds a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad f(x) < f(x_0) + \varepsilon.$$

We note that f is continuous at x_0 if and only if it is both lower and upper semi-continuous at that point.

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本页继续“实值半连续的定义与等价判据”。完整定义、条件和论证见本讲原页第 23 页下方。

25 原课件第 25 页：实值半连续的定义与等价判据

Numerical Functions Defined on a Topological Space

Properties of upper semi-continuous functions can easily be deduced from those of lower semi-continuous functions by replacing $f(x)$ by $-f(x)$; we shall therefore consider only the case of lower semi-continuous functions.

We say that a function f is lower semi-continuous (l.s.c.) (in X) if it is lower semi-continuous at each point of X.

We say that a function f is upper semi-continuous (u.s.c.) (in X) if it is upper semi-continuous at each point of X.

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本页继续“实值半连续的定义与等价判据”。完整定义、条件和论证见本讲原页第 23 页下方。

26 原课件第 26 页：实值半连续的定义与等价判据

Numerical Functions Defined on a Topological Space

Theorem 1. A necessary and sufficient condition for a numerical function f to be lower semi-continuous is that, for each number λ, the set $S_\lambda^- = \{x \in X \mid f(x) > \lambda\}$ is open.

Proof. The family $\mathscr{D}^-$ of intervals

$$\{(\lambda, +\infty) \mid \lambda \in \mathbb{R}\}$$

determines a topology in $\mathbb{R}$. Then it follows quickly from the definition of lower semi-continuity that f is lower semi-continuous if and only if it is a continuous mapping of X into the topological space $(\mathbb{R}, \mathscr{D}^-)$. Since $S_\lambda^- = f^{-1}((\lambda, +\infty))$, the theorem stated is equivalent to this result. □

Corollary 1. A numerical function f is lower semi-continuous if and only if, for each number λ, the lower contour set $T_\lambda^- = \{x \mid f(x) \leqq \lambda\}$ is closed.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“实值半连续的定义与等价判据”。完整定义、条件和论证见本讲原页第 23 页下方。

27 原课件第 27 页：实值半连续的定义与等价判据

Numerical Functions Defined on a Topological Space

Theorem 1. A necessary and sufficient condition for a numerical function f to be upper semi-continuous is that, for each number λ, the set $S_\lambda^+ = \{x \in X \mid f(x) < \lambda\}$ is open.

Proof. The family $\mathscr{D}^+$ of intervals

$$\{(-\infty, \lambda) \mid \lambda \in \mathbb{R}\}$$

determines a topology in $\mathbb{R}$. Then it follows quickly from the definition of upper semi-continuity that f is upper semi-continuous if and only if it is a continuous mapping of X into the topological space $(\mathbb{R}, \mathscr{D}^+)$. Since $S_\lambda^+ = f^{-1}((-\infty, \lambda))$, the theorem stated is equivalent to this result. □

Corollary 1. A numerical function f is upper semi-continuous if and only if, for each number λ, the upper contour set $T_\lambda^+ = \{x \mid f(x) \geqq \lambda\}$ is closed.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“实值半连续的定义与等价判据”。完整定义、条件和论证见本讲原页第 23 页下方。

28 原课件第 28 页：实值半连续的定义与等价判据

Numerical Functions Defined on a Topological Space

Theorem 1. A necessary and sufficient condition for a numerical function f to be lower semi-continuous is that, for each number λ, the set $S_\lambda^- = \{x \in X \mid f(x) > \lambda\}$ is open, i.e. the lower contour set $T_\lambda^- = \mathbf{C}_X S_\lambda^- = \{x \mid f(x) \leqq \lambda\}$ is closed.

Proof. If the set $S_\lambda^- = \{x \in X \mid f(x) > \lambda\}$ is an open set for every $\lambda \in \mathbb{R}$, consider any $x_0 \in X$, then for any $\epsilon > 0$, $S_{f(x_0)-\epsilon}^- = \{x \in X \mid f(x) > f(x_0) - \epsilon\}$ is an open set and $x \in S_{f(x_0)-\epsilon}^-$, then there exists a neighborhood U of x_0 such that $U \subseteq S_{f(x_0)-\epsilon}^-$, then $f(x) > f(x_0) - \epsilon$ for all $x \in U$. Hence f is lower semi-continuous.

Conversely, if $f : X \to \mathbb{R}$ is lower semi-continuous, then for any $\lambda \in \mathbb{R}$, let $S_\lambda^- = \{x \in X \mid f(x) > \lambda\}$. For any $x_0 \in S_\lambda^-$, $f(x_0) > \lambda$, then let $\epsilon = \dfrac{f(x_0) - \lambda}{2}$, there exists a neighborhood U of x_0 such that $f(x_0) - \epsilon < f(x)$ for all $x \in U$, as $f(x_0) - \epsilon = \dfrac{f(x_0) + \lambda}{2} > \lambda$, $U \subseteq S_\lambda^-$, then S_λ^- is open. $\square$

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本页继续“实值半连续的定义与等价判据”。完整定义、条件和论证见本讲原页第 23 页下方。

29 原课件第 29 页：实值半连续的定义与等价判据

Numerical Functions Defined on a Topological Space

Theorem 1. A necessary and sufficient condition for a numerical function f to be upper semi-continuous is that, for each number λ, the set $S_\lambda^+ = \{x \in X \mid f(x) < \lambda\}$ is open, i.e. the upper contour set $T_\lambda^+ = \mathbf{C}_X S_\lambda^+ = \{x \mid f(x) \geqq \lambda\}$ is closed.

Proof. If the set $S_\lambda^+ = \{x \in X \mid f(x) < \lambda\}$ is an open set for every $\lambda \in \mathbb{R}$, consider any $x_0 \in X$, then for any $\epsilon > 0$, $S_{f(x_0)+\epsilon}^+ = \{x \in X \mid f(x) < f(x_0) + \epsilon\}$ is an open set and $x \in S_{f(x_0)+\epsilon}^+$, then there exists a neighborhood U of x_0 such that $U \subseteq S_{f(x_0)+\epsilon}^+$, then $f(x) < f(x_0) + \epsilon$ for all $x \in U$. Hence f is upper semi-continuous.

Conversely, if $f : X \to \mathbb{R}$ is upper semi-continuous, then for any $\lambda \in \mathbb{R}$, let $S_\lambda^+ = \{x \in X \mid f(x) < \lambda\}$. For any $x_0 \in S_\lambda^+$, $f(x_0) < \lambda$, then let $\epsilon = \dfrac{\lambda - f(x_0)}{2}$, there exists a neighborhood U of x_0 such that $f(x) < f(x_0) + \epsilon$ for all $x \in U$, as $f(x_0) + \epsilon = \dfrac{f(x_0) + \lambda}{2} < \lambda$, $U \subseteq S_\lambda^+$, then S_λ^+ is open. □

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本页继续“实值半连续的定义与等价判据”。完整定义、条件和论证见本讲原页第 23 页下方。

30 原课件第 30 页：半连续函数的极值证明

Numerical Functions Defined on a Topological Space

Theorem 2. If K is a compact subset of X, a lower semi-continuous function f attains in K the value $m = \inf_{x \in K} f(x)$ (in other words, $\inf_{x \in K} f(x)$ is a minimum).

Proof. Let μ be a number such that $\mu > m$. Then $S_\mu^- = \{x \in K \mid f(x) \leqq \mu\}$ is not empty and is a closed subset of the compact set K; hence, by the finite intersection axiom, we have

$$\bigcap_{\mu > m} S_\mu^- \neq \varnothing.$$

For any point x_0 of this intersection, we have $f(x_0) = m$. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

非空紧 K 上有限实值下半连续函数先必有下界：若无，取 $f(x_n) < -n$，紧性给子列趋于 $x \in K$，下半连续将推出有限 $f(x) \leq -\infty$，矛盾。设有限 $m = \inf_K f$，选 $f(x_n) < m + 1/n$，再取收敛子列，得到 $f(x) \leq \liminf f(x_n) = m$，而下界给反向，故取到最小值。对 $-f$ 应用此结论得上半连续函数有最大值。原页35趋近的值应为 M 而非 m。不能只从紧集直接说任意函数有界。

31 原课件第 31 页：半连续函数的极值证明

Numerical Functions Defined on a Topological Space

Theorem 2. Let f be a lower semi-continuous function on a compact metric space X. Then f is bounded below and attains its minimum.

Proof. Let $U_\lambda^- = (\lambda, +\infty)$. $\left\{U_\lambda^-; \lambda \in \Lambda\right\}$ be an open covering of $f(X)$. Then the sets $\left\{f^{-1}\left(U_\lambda^-\right); \lambda \in \Lambda\right\}$ form an open covering of X since we have show

$$f^{-1}\left(\bigcup_{\lambda\in\Lambda} U_\lambda^-\right) = \bigcup_{\lambda\in\Lambda} f^{-1}\left(U_\lambda^-\right).$$

Since X is compact, there exists a finite subset $\Lambda' = \{\lambda_1, \cdots, \lambda_N\}$ such that $\bigcup_{\lambda\in\Lambda'} f^{-1}\left(U_\lambda^-\right)$ still forms a covering of X, then we have

$$f(X) = f\left(\bigcup_{\lambda\in\Lambda'} f^{-1}\left(U_\lambda^-\right)\right) = f\left(f^{-1}\left(\bigcup_{\lambda\in\Lambda'} U_\lambda^-\right)\right) \subseteq \bigcup_{\lambda\in\Lambda'} U_\lambda^-$$

hence $f(x) \geqq \min\{\lambda_1, \cdots, \lambda_N\}$ for $x \in X$. Then f is bounded below.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“半连续函数的极值证明”。完整定义、条件和论证见本讲原页第 30 页下方。

32 原课件第 32 页：半连续函数的极值证明

Numerical Functions Defined on a Topological Space

Let $m = \inf_{x \in X} f(x)$, there exists a sequence $x_n \in X$ such that $f(x_n) \to m$. Since $x_n \in X$ and X is compact, we have some $x_0 \in X$ and some subsequence such that $x_{n_k} \to x_0$ as X is compact, and hence sequentially compact. As f is lower semi-continuous, for every $\epsilon > 0$ there exists a neighborhood U of x_0 such that $f(x_0) - \epsilon < f(x)$ for all $x \in U$, as $x_{n_k} \to x_0$, there exists $K \in \mathbb{N}$ such that for all $k \geqq K$, $x_{n_k} \in U$, hence

$$f(x_0) - \epsilon < f(x_{n_k})$$

let $k \to \infty$, we have

$$f(x_0) - \epsilon \leqq m.$$

As ϵ is arbitrary, we have $f(x_0) \leqq m$, and by definition of m, we have $f(x_0) \geqq m$, hence x_0 is a minimizer. □

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本页继续“半连续函数的极值证明”。完整定义、条件和论证见本讲原页第 30 页下方。

33 原课件第 33 页：半连续函数的极值证明

Numerical Functions Defined on a Topological Space

Theorem 2. If K is a compact subset of X, a upper semi-continuous function f attains in K the value $M = \sup_{x \in K} f(x)$ (in other words, $\sup_{x \in K} f(x)$ is a maximum).

Proof. Let μ be a number such that $\mu < M$. Then $S_\mu^+ = \{x \in K \mid f(x) \geqq \mu\}$ is not empty and is a closed subset of the compact set K; hence, by the finite intersection axiom, we have

$$\bigcap_{\mu < M} S_\mu^+ \neq \varnothing.$$

For any point x_0 of this intersection, we have $f(x_0) = M$. □

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34 原课件第 34 页：半连续函数的极值证明

Numerical Functions Defined on a Topological Space

Theorem 2. Let f be a upper semi-continuous function on a compact metric space X. Then f is bounded above and attains its maximum.

Proof. Let $U_\lambda^+ = (-\infty, \lambda)$. $\left\{U_\lambda^+; \lambda \in \Lambda\right\}$ be an open covering of $f(X)$. Then the sets $\left\{f^{-1}\left(U_\lambda^+\right); \lambda \in \Lambda\right\}$ form an open covering of X since we have show

$$f^{-1}\left(\bigcup_{\lambda\in\Lambda} U_\lambda^+\right) = \bigcup_{\lambda\in\Lambda} f^{-1}\left(U_\lambda^+\right).$$

Since X is compact, there exists a finite subset $\Lambda' = \{\lambda_1, \cdots, \lambda_N\}$ such that $\bigcup_{\lambda\in\Lambda'} f^{-1}\left(U_\lambda^+\right)$ still forms a covering of X, then we have

$$f(X) = f\left(\bigcup_{\lambda\in\Lambda'} f^{-1}\left(U_\lambda^+\right)\right) = f\left(f^{-1}\left(\bigcup_{\lambda\in\Lambda'} U_\lambda^+\right)\right) \subseteq \bigcup_{\lambda\in\Lambda'} U_\lambda^+$$

hence $f(x) \leqq \max\{\lambda_1, \cdots, \lambda_N\}$ for $x \in X$. Then f is bounded above.

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35 原课件第 35 页：半连续函数的极值证明

Numerical Functions Defined on a Topological Space

Let $M = \sup_{x \in X} f(x)$, there exists a sequence $x_n \in X$ such that $f(x_n) \to m$. Since $x_n \in X$ and X is compact, we have some $x_0 \in X$ and some subsequence such that $x_{n_k} \to x_0$ as X is compact, and hence sequentially compact. As f is upper semi-continuous, for every $\epsilon > 0$ there exists a neighborhood U of x_0 such that $f(x) < f(x_0) + \epsilon$ for all $x \in U$, as $x_{n_k} \to x_0$, there exists $K \in \mathbb{N}$ such that for all $k \geqq K$, $x_{n_k} \in U$, hence

$$f(x_{n_k}) < f(x_0) + \epsilon$$

let $k \to \infty$, we have

$$M \leqq f(x_0) + \epsilon.$$

As ϵ is arbitrary, we have $M \leqq f(x_0)$, and by definition of M, we have $f(x_0) \leqq M$, hence x_0 is a maximizer. $\square$

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36 原课件第 36 页：上确界、下确界与有限运算

Numerical Functions Defined on a Topological Space

Theorem 3. If $\{f_i \mid i \in I\}$ is a family of lower semi-continuous functions, the function g such that $g(x) = \sup_{i \in I} f_i(x)$ is lower semi-continuous.

Proof. Let x_0 be any point of X. Given $\varepsilon > 0$, there exists an index i such that

$$f_i(x_0) > g(x_0) - \varepsilon.$$

Since f_i is lower semi-continuous, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad f_i(x) > f_i(x_0) - \varepsilon > g(x_0) - 2\varepsilon$$

and therefore

$$x \in U(x_0) \quad \Longrightarrow \quad g(x) \geqq g(x_0) - 2\varepsilon.$$

Hence g is lower semi-continuous at x_0; since x_0 is arbitrary, the theorem is proved. □

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若 $g = \sup_i f_i$ 且各 f_i 下半连续，则 $\{g > a\} = \bigcup_i \{f_i > a\}$ 开，故 g 下半连续。上半连续函数的任意下确界对偶。有限下半连续函数的最小值满足 $\{\min_i f_i > a\} = \bigcap_i \{f_i > a\}$，有限交开；上半连续函数有限最大值对偶。无限交不保证开，故不能删掉“有限”。若确界可能无穷，应注明把函数视为扩展实值。

37 原课件第 37 页：上确界、下确界与有限运算

Numerical Functions Defined on a Topological Space

Theorem 3. If $\{f_i \mid i \in I\}$ is a family of upper semi-continuous functions, the function g such that $g(x) = \inf_{i \in I} f_i(x)$ is upper semi-continuous.

Proof. Let x_0 be any point of X. Given $\varepsilon > 0$, there exists an index i such that

$$f_i(x_0) < g(x_0) + \varepsilon.$$

Since f_i is upper semi-continuous, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \implies f_i(x) < f_i(x_0) + \varepsilon < g(x_0) + 2\varepsilon$$

and therefore

$$x \in U(x_0) \implies g(x) \leqq g(x_0) + 2\varepsilon.$$

Hence g is upper semi-continuous at x_0; since x_0 is arbitrary, the theorem is proved. □

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本页继续“上确界、下确界与有限运算”。完整定义、条件和论证见本讲原页第 36 页下方。

38 原课件第 38 页：上确界、下确界与有限运算

Numerical Functions Defined on a Topological Space

Theorem 4. If $f_1, \cdots, f_N$ are lower semi-continuous functions then the function $g(x) = \inf_{n \in I} f_n(x)$ (where $I = \{1, \cdots, N\}$) is lower semi-continuous.

Proof. Let x_0 be any point of X. Given $\varepsilon > 0$, there exists a neighbourhood $U_n(x_0)$ such that

$$x \in U_n(x_0) \quad \Longrightarrow \quad f_n(x) > f_n(x_0) - \varepsilon.$$

If $x \in \bigcap_{n \in I} U_n(x_0)$, then, for all n,

$$f_n(x) > f_n(x_0) - \varepsilon,$$

whence

$$g(x) = \inf_{n \in I} f_n(x) \geqq \inf_{n \in I} \{f_n(x_0) - \varepsilon\} = g(x_0) - \varepsilon$$

and so g is lower semi-continuous. □

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39 原课件第 39 页：上确界、下确界与有限运算

Numerical Functions Defined on a Topological Space

Theorem 4. If $f_1, \cdots, f_N$ are upper semi-continuous functions then the function $g(x) = \sup_{n \in I} f_n(x)$ (where $I = \{1, \cdots, N\}$) is upper semi-continuous.

Proof. Let x_0 be any point of X. Given $\varepsilon > 0$, there exists a neighbourhood $U_n(x_0)$ such that

$$x \in U_n(x_0) \quad \Longrightarrow \quad f_n(x) < f_n(x_0) + \varepsilon.$$

If $x \in \bigcap_{n \in I} U_n(x_0)$, then, for all n,

$$f_n(x) < f_n(x_0) + \varepsilon,$$

whence

$$g(x) = \sup_{n \in I} f_n(x) \leqq \sup_{n \in I} \{f_n(x_0) + \varepsilon\} = g(x_0) + \varepsilon$$

and so g is upper semi-continuous. □

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40 原课件第 40 页：半连续函数的加法与正乘法

Numerical Functions Defined on a Topological Space

Theorem 5. If f_1 and f_2 are two lower semi-continuous numerical functions, the function $f_1 + f_2$ is lower semi-continuous.

Proof. The set

$$\{x \mid f_1(x) + f_2(x) > a\} = \bigcup_{\lambda} \{x \mid f_1(x) > a - \lambda\} \cap \{x \mid f_2(x) > \lambda\}$$

is a union of open sets and so is open. □

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下半连续 f, g 的和下半连续：给定 ε，分别取邻域使 $f(z) > f(x) - \varepsilon/2$ 和 $g(z) > g(x) - \varepsilon/2$，在交邻域相加。对上半连续用负号对偶。正函数的乘积可用单调连续的 $\log$：$\log f, \log g$ 保持相应半连续性，和半连续，再用单调连续的指数还原。也可直接给乘积误差界。正性不能无故删掉，因为乘负数会反转半连续的方向。

41 原课件第 41 页：半连续函数的加法与正乘法

Numerical Functions Defined on a Topological Space

Theorem 5. If f_1 and f_2 are two upper semi-continuous numerical functions, the function $f_1 + f_2$ is upper semi-continuous.

Proof. The set

$$\{x \mid f_1(x) + f_2(x) < a\} = \bigcup_{\lambda} \{x \mid f_1(x) < a - \lambda\} \cap \{x \mid f_2(x) < \lambda\}$$

is a union of open sets and so is open. □

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42 原课件第 42 页：半连续函数的加法与正乘法

Numerical Functions Defined on a Topological Space

Theorem 6. If f_1 and f_2 are two positive lower semi-continuous numerical functions, the function h defined by $h(x) = f_1(x)f_2(x)$ is lower semi-continuous.

Proof. If $\alpha > 0$, the set

$$\{x \mid f_1(x)f_2(x) > \alpha\} = \bigcup_{\lambda>0} \{x \mid f_1(x) > \lambda\} \cap \left\{x \mid f_2(x) > \frac{\alpha}{\lambda}\right\}$$

is a union of open sets and so is open. □

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43 原课件第 43 页：半连续函数的加法与正乘法

Numerical Functions Defined on a Topological Space

Theorem 6. If f_1 and f_2 are two positive upper semi-continuous numerical functions, the function h defined by $h(x) = f_1(x)f_2(x)$ is upper semi-continuous.

Proof. If $\alpha > 0$, the set

$$\{x \mid f_1(x)f_2(x) < \alpha\} = \bigcup_{\lambda>0} \{x \mid f_1(x) < \lambda\} \cap \left\{x \mid f_2(x) < \frac{\alpha}{\lambda}\right\}$$

is a union of open sets and so is open. □

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